

Navroop Singh

Systems Engineer

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SHEET	TITLE	ISSUED	STATUS
R.01	Résumé	Aug 2026	Open to rôles

PROFILE

Systems engineer with a mechanical engineering foundation and two years of industrial commissioning experience in a polysilicon process plant, now an M.Sc. Systems Engineering student at Otto von Guericke University Magdeburg. Work spans rotating equipment, maintenance engineering and P&ID / GAD interpretation through to equipment installation, commissioning and plant optimization. Moving toward MBSE, system architecture and simulation-driven production systems.

Availability: up to 20 h/week during semester, full-time in breaks · **Earliest start:** immediate · **Location:** Magdeburg, open to relocating anywhere in Germany and to remote work · **Work authorisation:** non-EU student, eligible for working-student employment under German student visa terms.

EXPERIENCE

Aug 2024 – Sep 2025 Jamnagar, India	Rotating Equipment Engineer — Installation & Commissioning Reliance Corp IT Park Ltd (Reliance Industries — New Energy) — Polysilicon Plant · designation: Manager - Supervised installation of three reciprocating compressors — two horizontal, one vertical — covering vendor coordination, site sequencing and quality inspection through to pre-commissioning. - Witnessed the Mechanical Run Test at the vendor's works for one compressor, and one canned seal-less pump tag on site. - Reviewed and interpreted General Arrangement Drawings (GAD) and P&IDs for rotating equipment installations. - Supervised nitrogen preservation procedures ahead of commissioning. - Developed a reactor bell shifting and sequencing plan projected to remove ~14 h of downtime per batch (modelled — plant in construction phase).
Aug 2023 – Aug 2024 Jamnagar, India	Graduate Engineering Trainee Reliance Corp IT Park Ltd (Reliance Industries — New Energy) — Polysilicon Plant - Assisted the rotating equipment team during installation, assembly and pre-commissioning. - Supported coordination between maintenance teams, vendors and project engineers. - Participated in documentation reviews, quality inspections and MRT observations.
Dec 2021 – Jun 2022 India	R&D Intern Escorts Kubota Limited — Agricultural & Construction Machinery - Built an Excel/VBA kinematics and force-analysis template for backhoe loader linkages, automating geometry and free-body calculations previously done by hand. - Cut manual calculation time for linkage kinematic analysis by ~30%. - Modelled a quick-hitch coupler in Siemens Solid Edge. - Internship formally certified by Escorts Kubota HR (EAM & ECE – R&D).

EDUCATION

2025 – Expected 2027 Magdeburg, DE	M.Sc. Systems Engineering for Manufacturing Otto von Guericke University Magdeburg Systems Engineering · MBSE · System Architecture · Systems Modeling (SysML, UML) · Control Systems · Systems Integration · Modelling & Simulation of Logistics Planning · Python in Production Systems · Engineering of Resilient Production Systems · Human-Technology Interaction
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SELECTED PROJECTS

Backhoe Loader Kinematics & Force Analysis ESCORTS KUBOTA

Excel/VBA kinematics and force-analysis template for backhoe loader linkages — linkage geometry driving a full free-body solution at every pin joint, with computed reach, working height, bucket rotation and breakout force. Cut manual calculation time by ~30%. Numeric values withheld — Escorts Kubota internal data

PPR Modelling & Validation Tool GROUP OF THREE

Interactive web app modelling Product-Process-Resource relationships as a directed graph — import from JSON, XML or Excel, in-place editing with PPR conformance enforced at insertion time, two engineering views (basic engineering and reliability) over the shared graph, and four requirement-check algorithms. Deployed on Streamlit Cloud. Registered as a group of three; built by two of us. My scope: the engineering and reliability views. Python · Streamlit · NetworkX · Graphviz · PyVis

Production Scheduling & Lot-Sizing Study SUMMER SEMESTER 2026

Modelled a six-product, five-stage line (Press → Laser → Bending → Profiling → Welding, two-machine Profiling bottleneck) in Tecnomatix Plant Simulation and benchmarked five order-release strategies over 100 orders / 1,050 units against makespan, total tardiness and on-time delivery. A lot-based search cut makespan 20% and tardiness 64% against the family-production baseline, but EDD eliminated tardiness entirely and won the weighted objective — so the recommendation is lot-based release now, EDD after a SMED set-up-reduction programme. Tecnomatix Plant Simulation 2404 · SimTalk

Thermal Printer MBSE Model (RAMI 4.0) GROUP OF TWO

Modelled a thermal printer in Enterprise Architect against the RAMI 4.0 reference architecture — two hierarchy levels (Work Station, Control Device), the six RAMI layers plus a Product-Process-Resource layer, refined top-down with traceability from workstation goals to physical realisation. My scope: the Sheet Feed Controller in full — all seven layers, every diagram and its traceability to the workstation requirements — plus the Business, Function and Information layers at Work Station level.

Medical Exoskeleton TEAM LEAD

Led a five-member capstone team designing a low-cost lower-limb exoskeleton — sized to a 78 kg user and 150 Nm swing torque, load calculations and component sizing in aluminium 6061 (FOS 4) with 304 stainless steel shafts, eight parts checked in FEA. Delivered a working prototype at €300, fully reimbursed by the institute.

SKILLS

SYSTEMS ENGINEERING	MBSE · SysML · UML · System Architecture · RAMI 4.0
CAD & DESIGN	SolidWorks · PTC Creo · AutoCAD · Siemens Solid Edge · Navisworks
MODELLING & SIMULATION	ANSYS · Siemens Tecnomatix Plant Simulation · Enterprise Architect (Sparx)
PROGRAMMING	Python · MATLAB · SimTalk · Streamlit · NetworkX · Graphviz
PLANT & DOMAIN	Rotating equipment · Maintenance engineering · P&ID / GAD · Installation & commissioning · Plant optimization
STANDARDS	ISO 13482 · ISO 13485 · ISO 14001 (working familiarity)
LANGUAGES	English (C1) · German (A2) · Punjabi (native) · Hindi (fluent)

COMMUNITY

2025 – Present Magdeburg, DE	Volunteer Member — DEGis Otto von Guericke University Magdeburg Planning and running cultural events, and coordinating society activities across a multicultural team.
2025 – Present Magdeburg, DE	Member — ENACTUS Student entrepreneurship network Attended the ENACTUS training weekend in Heidelberg in 2025 — workshops on time management and on the startup accelerator programme.